Navigation Satellite Systems

GPS Receiver Carrier Tracking
MATLAB/Simulink Implementation

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1 – PLL and FLL Implementation

I implemented the Phase-Locked Loop (PLL) and Frequency-Locked Loop (FLL) discriminators and verified their correct behavior. The PLL estimates phase error, while the FLL estimates frequency error. Initial tests with zero error confirmed that output fluctuate around zero due to noise. When artificial frequency and phase errors were introduced, the discriminators responded correctly, confirming proper implementation.

2 – Signal and Noise Power Analysis

In this task, I analyzed signal and noise power at three stages: front-end, Doppler removal, and after the accumulate-and-dump block. Power was computed using mean$(I^2 + Q^2)$. Two scenarios were simulated: signal-only and noise-only.

Results showed that signal power remains constant before accumulation, while noise power also remains constant. After the accumulate-and-dump block, signal power increased approximately with the square of the integration time (PIT), while noise power increased linearly with PIT.

This demonstrates that coherent integration strengthens the signal much faster than noise, resulting in improved signal-to-noise ratio (SNR).

3 – PLL/FLL Performance Analysis

In this task, I evaluated the performance of PLL and FLL discriminators under different SNR conditions. The system was tested under a perfect-reference case, where phase and frequency errors were zero, so that any variation in outputs is due to noise.

I computed the standard deviation of PLL (in cycles) and FLL (in cycles per second) outputs. As SNR decreased from -18 dB to -28 dB, both PLL and FLL output became noisier, showing increased standard deviation.

I then increased the integration time (PIT) to compensate for degraded SNR. The results showed that increasing PIT reduced the discriminator noise and restored performance close to the nominal case.

This confirms that increasing coherent integration improves tracking performance by reducing noise effects, although it comes with a trade-off of slower response to dynamic changes.